

TOPIC 7: MULTIPLICATION AND DIVISION OF DIRECTED NUMBERS

1. Rules for Multiplication

- Positive $\times$ Positive = Positive (e.g. $(+3) \times (+4) = +12$)
- Negative $\times$ Negative = Positive (e.g. $(-3) \times (-4) = +12$)
- Positive $\times$ Negative = Negative (e.g. $(+3) \times (-4) = -12$)
- Negative $\times$ Positive = Negative (e.g. $(-3) \times (+4) = -12$)

Same signs $\rightarrow$ positive product; different signs $\rightarrow$ negative product.

2. Rules for Division

The same sign rules apply as for multiplication.

- $(+12) \div (+3) = +4$
- $(-12) \div (-3) = +4$
- $(+12) \div (-3) = -4$
- $(-12) \div (+3) = -4$

3. Practical Exercises

Practise with integers and simple decimals. Combine with addition/subtraction using BODMAS where needed.

4. Summary

When multiplying or dividing directed numbers: same signs give a positive result; different signs give a negative result. Always apply the sign rules carefully.

End of Topic 7